Lecture 15: Bonding in organic molecules-2

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Classroom: Teaching Center 401

Wednesday/Friday: 10:15-11:00; 11:10-11:55

2025/12/05

Major points of last lecture

- **Bonding in organic molecules**

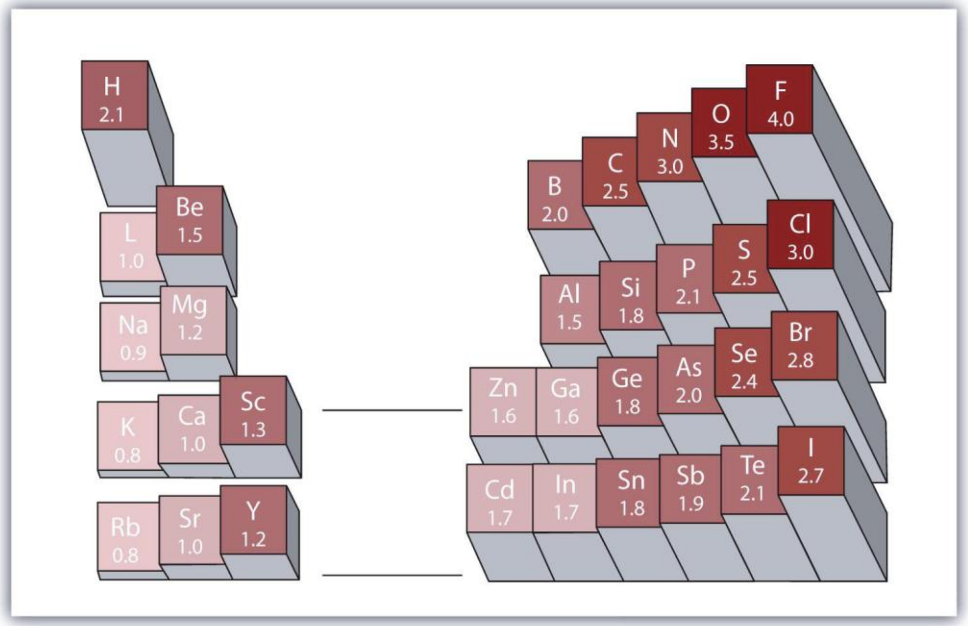
  1. Petroleum Refining and the Hydrocarbons
  2. The Alkanes
  3. The Alkenes and Alkynes
  4. Aromatic Hydrocarbons
  5. Fullerenes
  6. Functional Groups and Organic Reactions

Why Carbon?

Organic Chemistry:
The study of the compounds of carbon.

Average Bond Dissociation Energies ($kJ \cdot mol^{-1}$)

H—H	C—H	C—C	N—N	O—O	F—F	Si—O	P—O
436	414	347	161	146	153	460	377

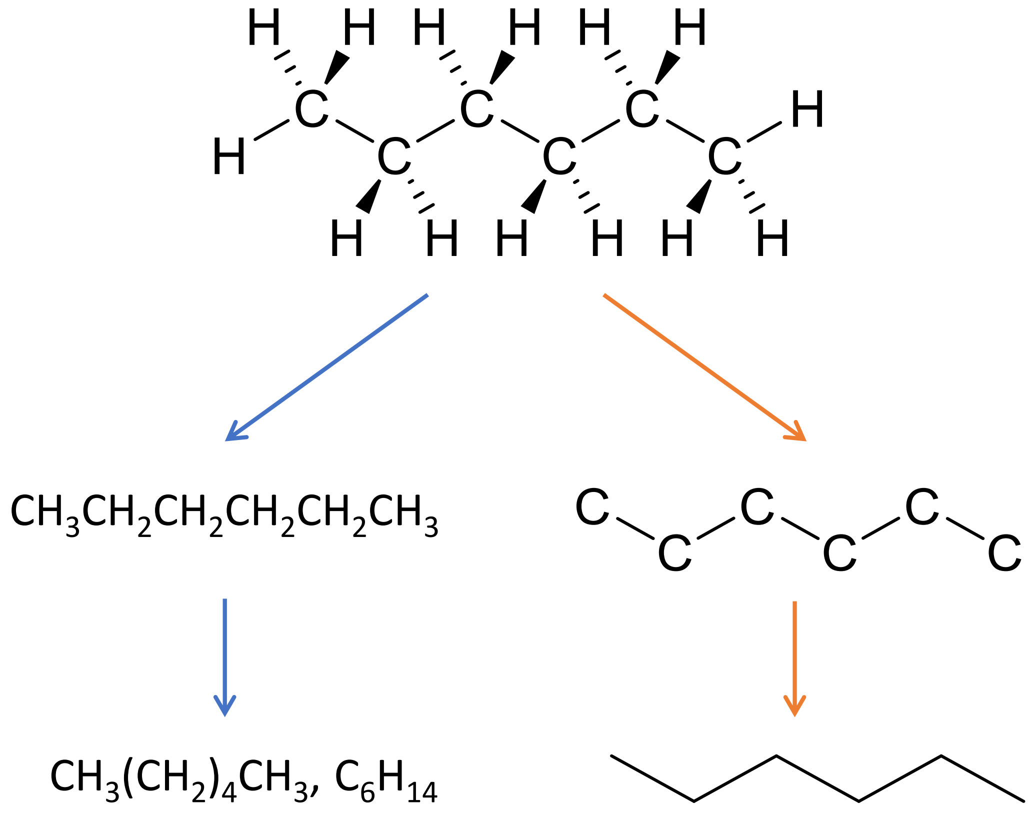


1. **Small: easily form the double and triple bonds**
2. **Can make four bonds and molecules with different geometries (linear, trigonal, tetrahedral)**
3. **Modest electronegativity: form covalent bonds with elements with both high and low electronegativity**

The Alkanes （烷烃）

The most prevalent **hydrocarbons** in petroleum are the **straight-chain alkanes**: chains of carbon atoms bonded to one another by single bonds, with enough hydrogen atoms on each carbon atom to bring it to the maximum bonding capacity of four.

Generic formula C_nH_{2n+2}

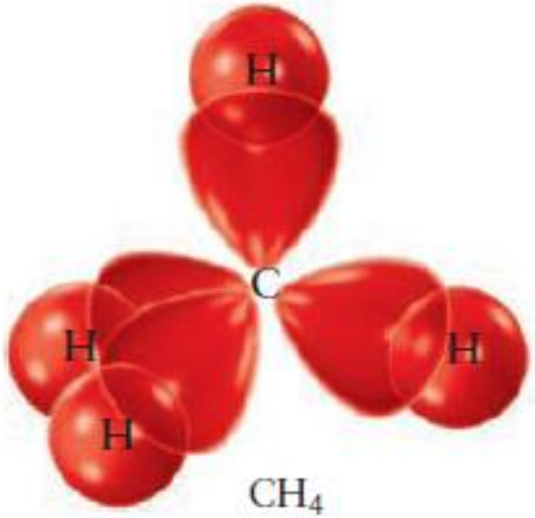


n	Name
1	Methane 甲烷
2	Ethane 乙烷
3	Propane 丙烷
4	Butane 丁烷
5	Pentane 戊烷
6	Hexane 己烷
7	Heptane 庚烷
8	Octane 辛烷
9	Nonane 壬烷
10	Decane 癸烷
11	Undecane 十一烷
12	Dodecane 十二烷
13	Tridecane 十三烷
14	Tetradecane 十四烷
15	Pentadecane 十五烷

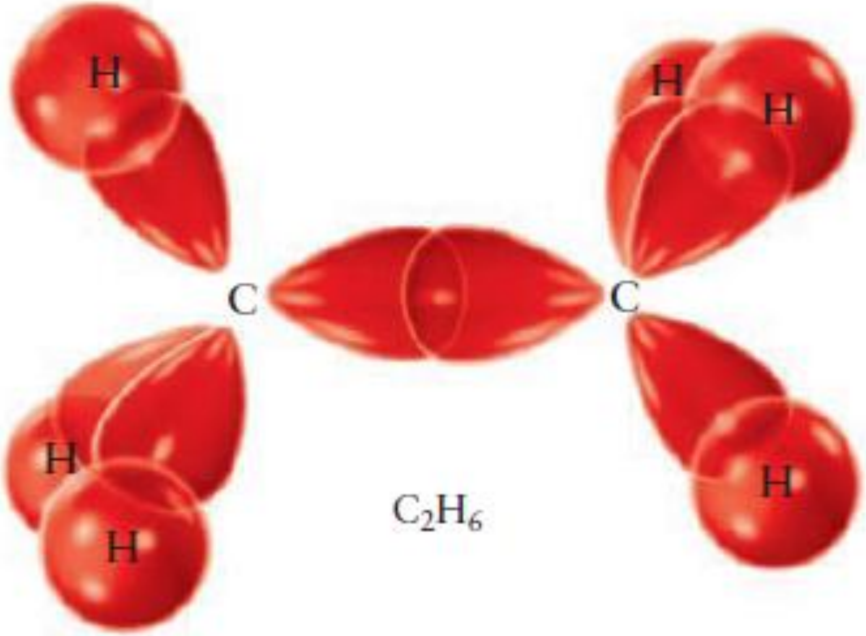
The Alkanes

C: $(1s)^2(2s)^2(2p)^2$ sp^3 hybridization

H: $(1s)^1$



CH_4



C_2H_6

The Alkanes

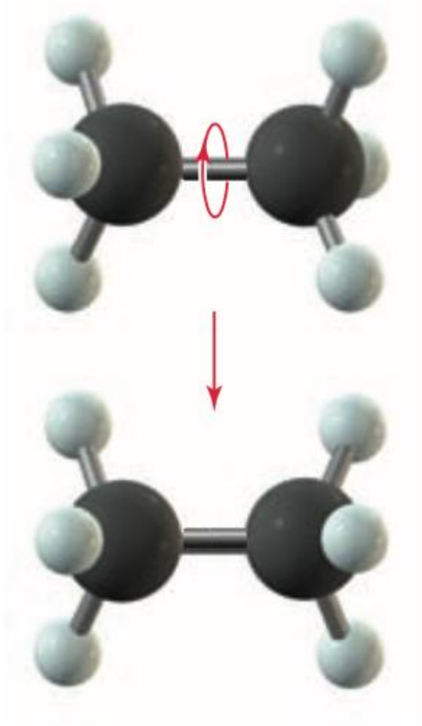
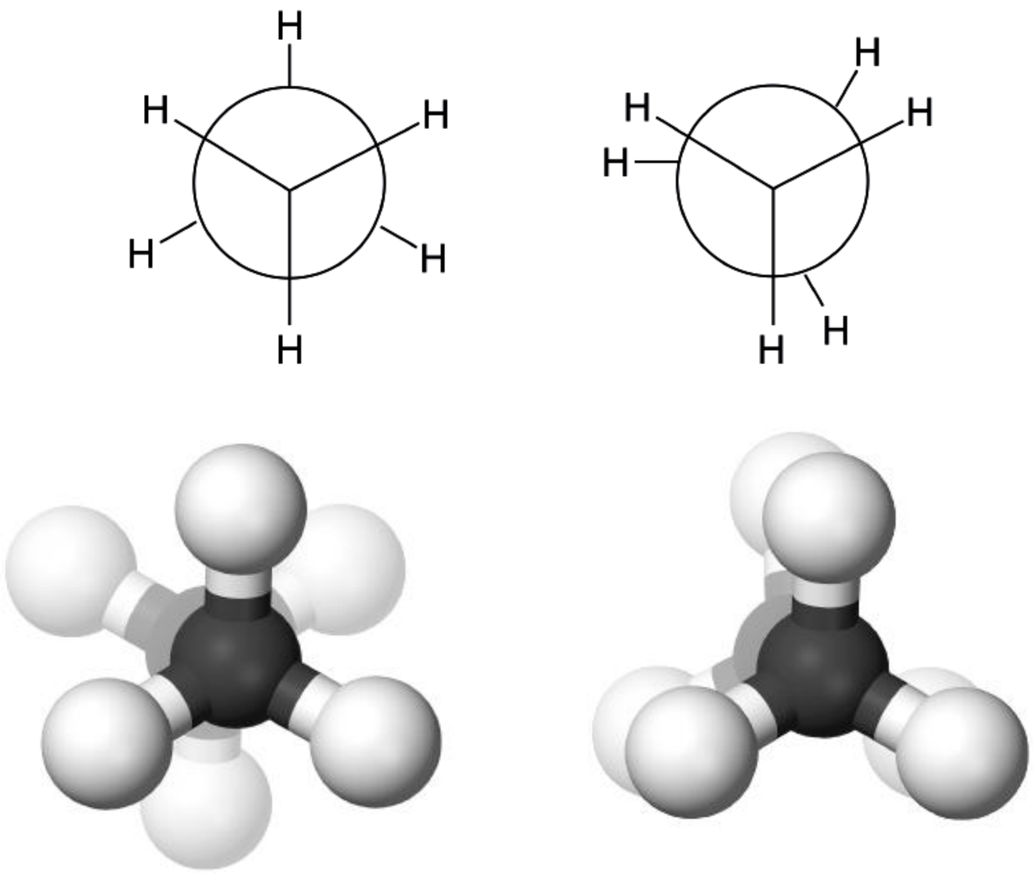


FIGURE 7.2 The two —CH_3 groups in ethane rotate easily about the bond that joins them.

different between σ bond and π bond

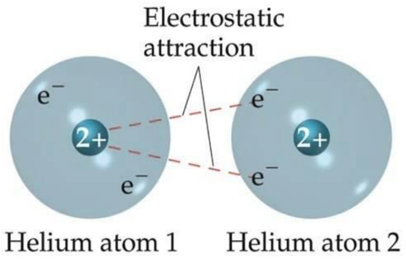
交错式(staggered) 重叠式(eclipsed)



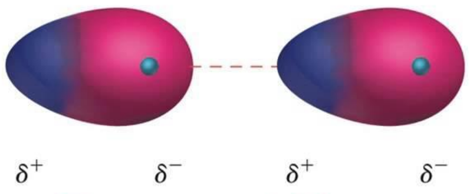
the staggered conformation is more stable by 12.5 kJ/mol than the eclipsed conformation.

The Alkanes

It is not a permanent partial charge



Even Distribution of electrons



Temporary UNeven distribution of e- which causes a temporary attraction

伦敦力（London dispersion forces），两个分子可以由于瞬时偶极（两个分子的瞬时偶极总是反向的）间的作用，产生引力，称为伦敦力

increasing strength of dispersion forces between heavier molecules

Butane C_4
B.p.沸点 ~0°C

Diesel C_{10}–C_{22}
B.p. 180–370°C

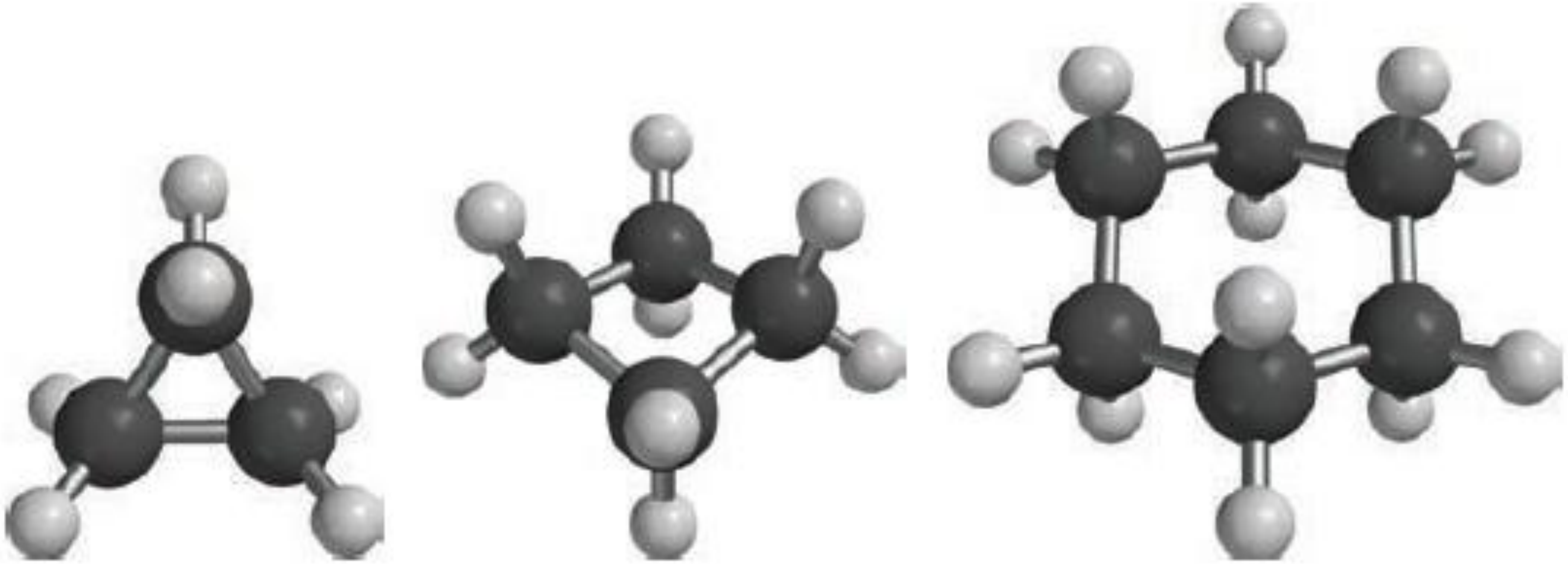
Vaseline C_{20}–C_{30}
M.p. ~37°C

HDPE $C_{\sim100000}$
M.p. ~130°C

Cyclic Alkanes （环烷烃）

A cycloalkane consists of at least one chain of carbon atoms attached at the ends to form a closed loop

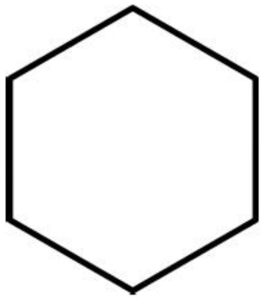
- cyclo- to the name of the straight-chain alkane
- general formula for cycloalkanes having one ring: C_nH_{2n}

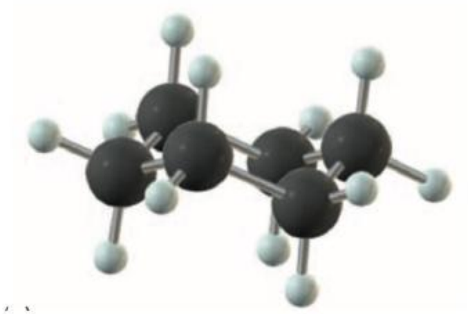


(a) (b) (c)

Cyclic Alkanes

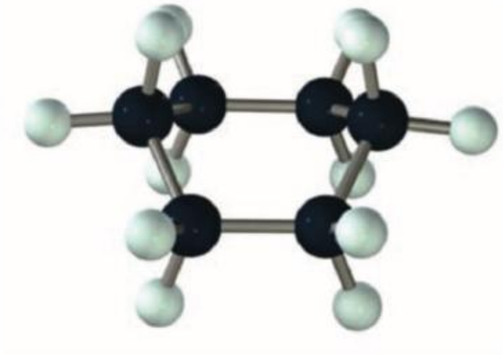
"tetrahedral angle of 109.5 degrees" + "a cyclic structure" = two new structural features (C_6H_{12})





chair conformation

椅型构象



boat conformation

船型构象

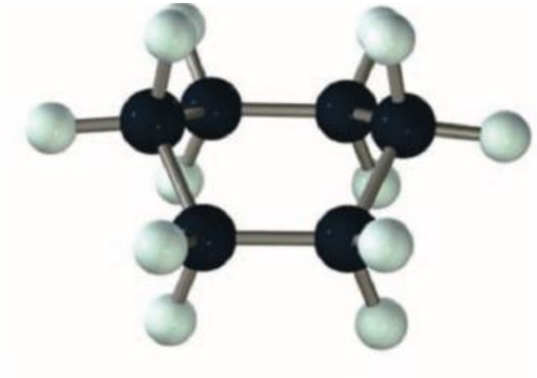
The chair conformation is more stable than the boat: the hydrogen atoms can become quite close and interfere with one another in the boat conformation.

Cyclic Alkanes

When atoms that are not bonded to each other come sufficiently close in space to experience a repulsive interaction, this increase in potential energy reduces the stability of the molecule. Such interactions are called steric strain (空间张力 或 位阻张力)

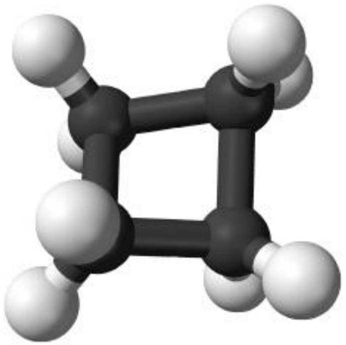
chair conformation

椅型构象



boat conformation

船型构象



This distortion of the bond angle from the tetrahedral value increases the potential energy of the bond above its stable equilibrium value, and the resulting angle strain (角应变)

The Alkanes：geometrical isomers几何异构体

(a)

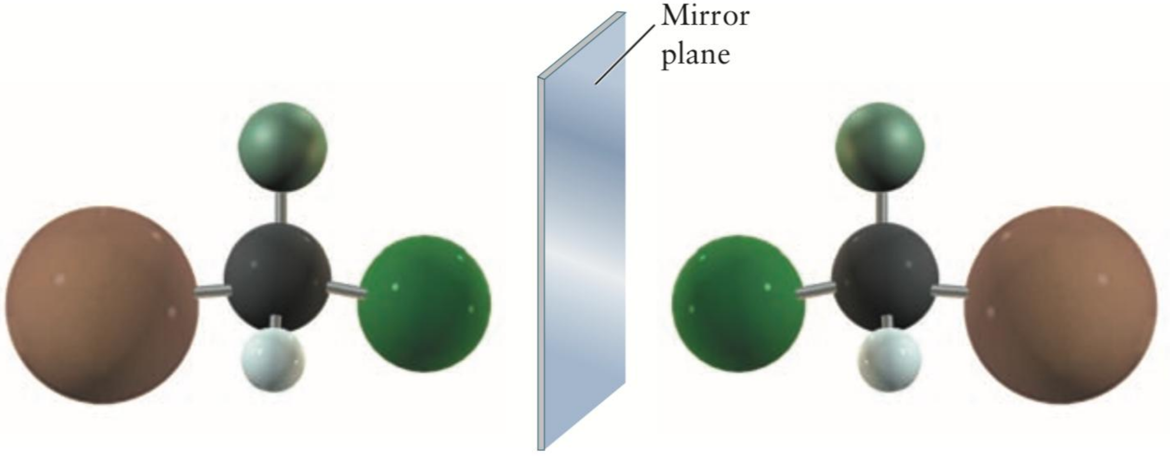
(b)

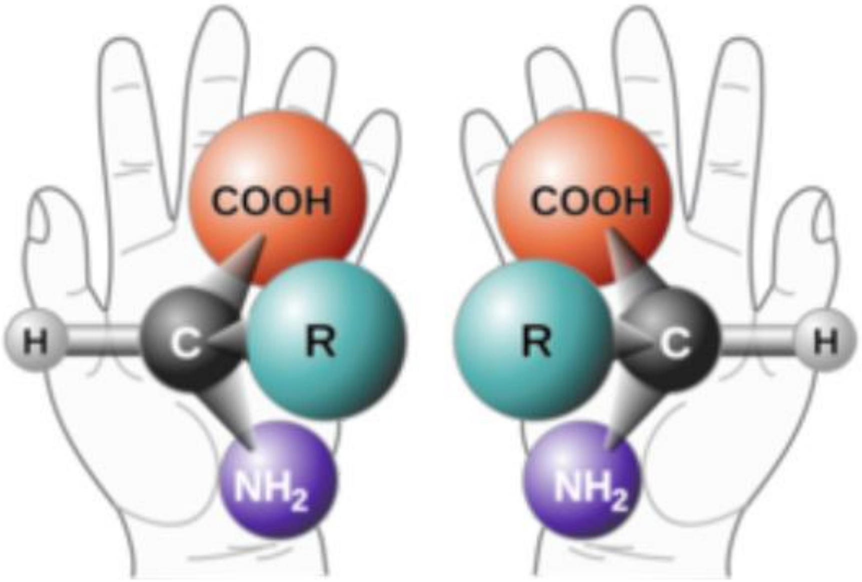
FIGURE 7.8 Two isomeric hydrocarbons with the molecular formula C_4H_{10}. (a) Butane. (b) 2-Methylpropane.

Geometrical isomers几何异构体: same molecular formula but a different bonding structure

The number of possible isomers increases rapidly with increasing numbers of carbon atoms in the hydrocarbon molecule.

The Alkanes：optical isomerism 光学异构体, chirality





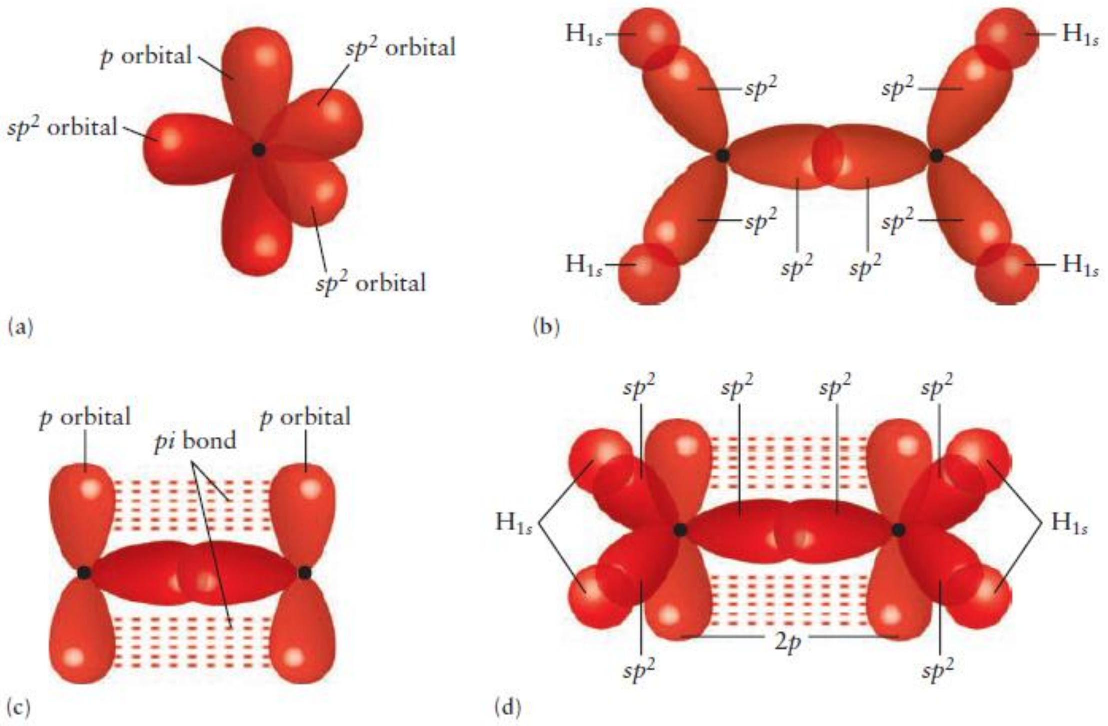
Optical isomerism 光学异构体**: the two forms rotate the plane of polarized light in different directions; therefore, such molecules are said to be "optically active".**

The Alkenes (烯烃) and Alkynes (炔烃)

- **Unsaturated hydrocarbons: have double and triple carbon-carbon bonds**

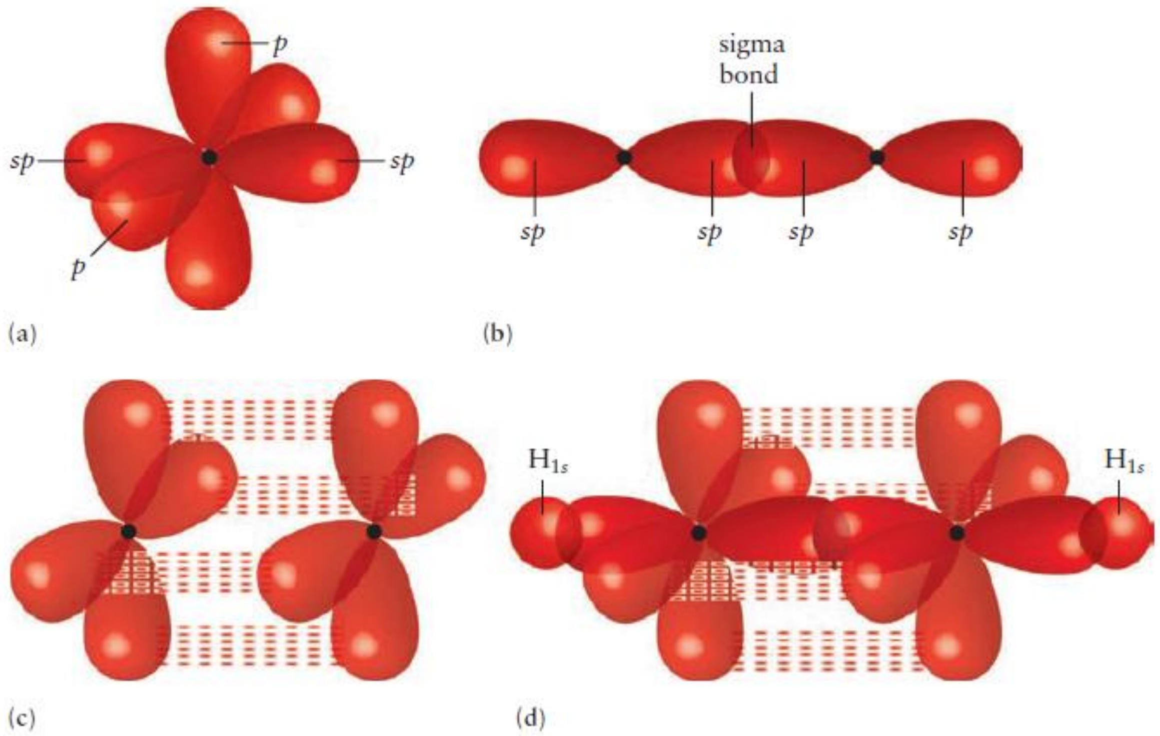
Ethylene (C_2H_4): a double bond between its carbon atoms and is called an alkene.

sp^2 hybridization



Ethylene (C_2H_2): a triple bond between its carbon atoms and is called an alkyne.

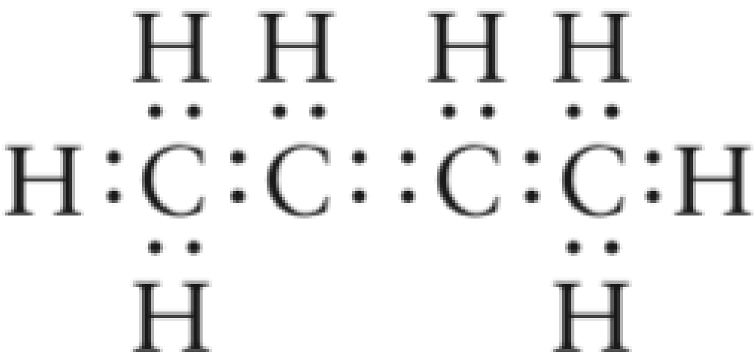
sp hybridization

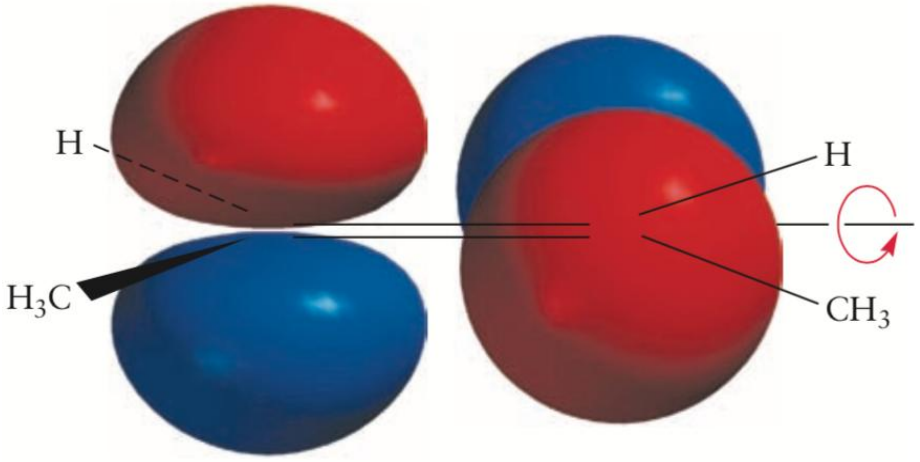


The Alkenes and Alkynes

Localized VB bonds to describe the molecular framework

Delocalized MOs to describe the π electrons



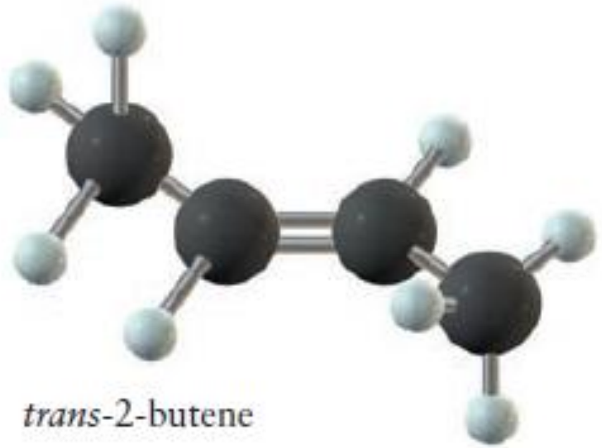


High-energy structure

- **The two outer carbon atoms: SN=4 (sp^3)**
- **The two central carbon atoms: SN=3 (sp^2)**
- **The p orbitals from the two central carbon atoms can be mixed to form a π_z (bonding) MO and a π_z* (antibonding) MO.**

The Alkenes and Alkynes

cis-2-butene



trans-2-butene

FIGURE 7.14 The two *cis–trans* isomers of 2-butene.

- **Breaking a π bond costs a significant amount of energy: both cis (顺式) and trans (反式) forms are stable, and interconversion between the two is slow.**

- **The change from trans to cis can be accomplished by photochemistry without complete breaking of bonds.**

- **Molecules such as trans-2-butene can absorb ultraviolet light, which excites an electron from a π to a π^* MO.**

The Alkenes and Alkynes

Compounds with two double bonds are called dienes (二烯类).

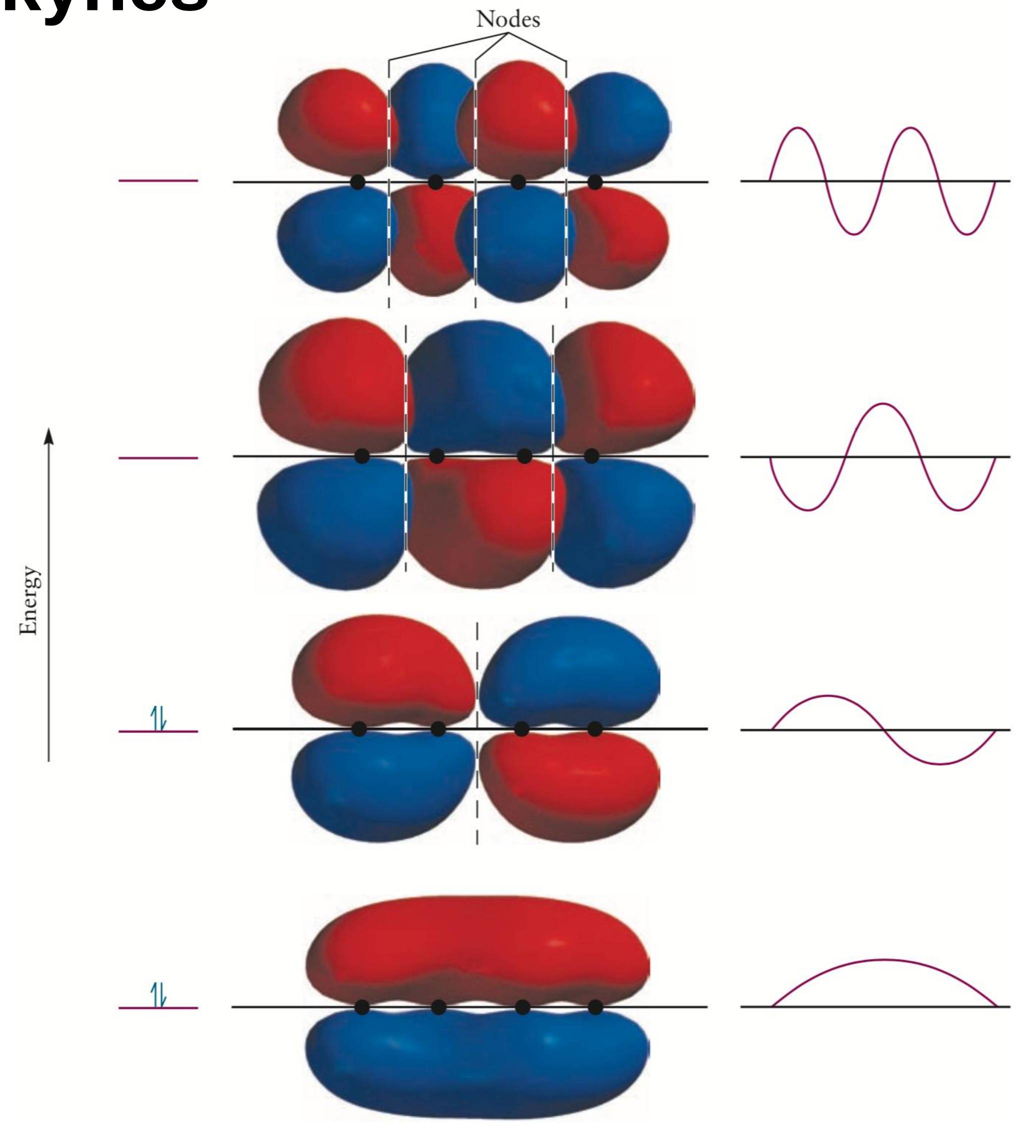
$CH_2{=}CH{-}CH{=}CH{-}CH_3$ **1,3-pentadiene (1,3-戊二烯)**

1,3-butadiene (1,3-丁二烯)

>2 double or triple bonds occur close to each other in a molecule: a delocalized MO.

- **All 4 carbon atoms have SN=3 (sp^2).**
- **The p_z orbitals have maximum overlap when the 4 carbon atoms lie in the same plane (predicted to be planar)**

The Alkenes and Alkynes

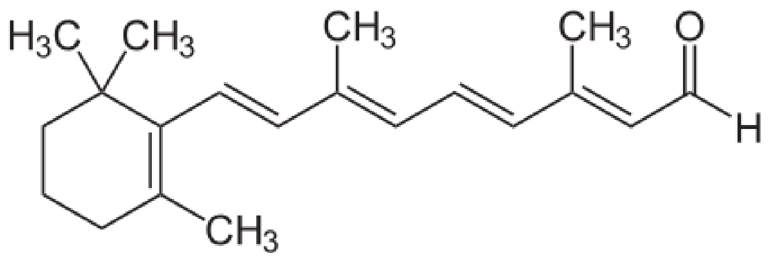


$CH_2{=}CH{-}CH{=}CH{-}CH_3$

one-dimensional particle-in-a-box

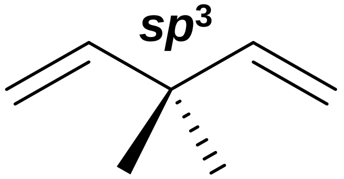
The Alkenes and Alkynes

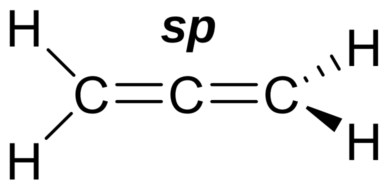
Conjugate system (共轭体系)

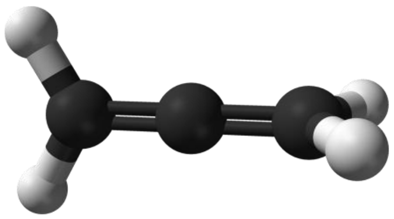


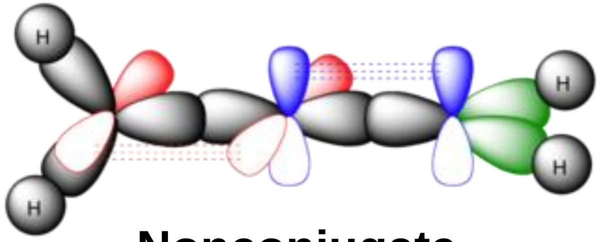
Retinal
视黄醛

Nonconjugate









Allene 丙二烯

Nonconjugate

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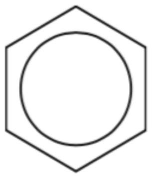
Major points of today's lecture

- **Bonding in organic molecules**

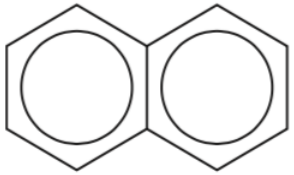
  1. **Petroleum Refining and the Hydrocarbons**
  2. **The Alkanes**
  3. **The Alkenes and Alkynes**
  4. **Aromatic Hydrocarbons**
  5. **Fullerenes**
  6. **Functional Groups and Organic Reactions**

Aromatic Hydrocarbons 芳香烃

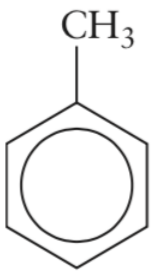
Aromatic compounds: contain benzene rings with various side groups or two (or more) benzene rings linked by alkyl chains or fused side by side, as in naphthalene ($C_{10}H_8$):



Benzene (苯)

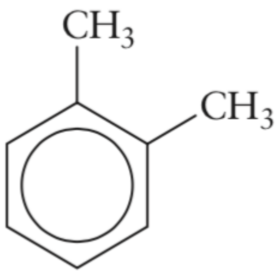


Naphthalene (萘)



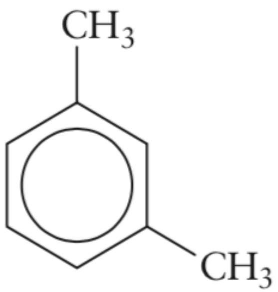
Toluene

甲苯



o-Xylene

邻二甲苯

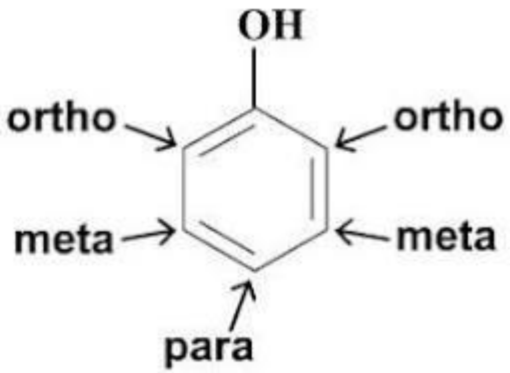


m-Xylene

间二甲苯

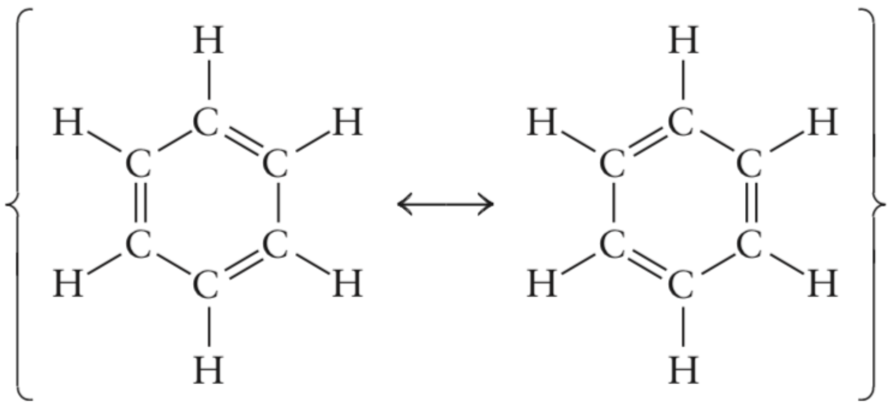
p-Xylene

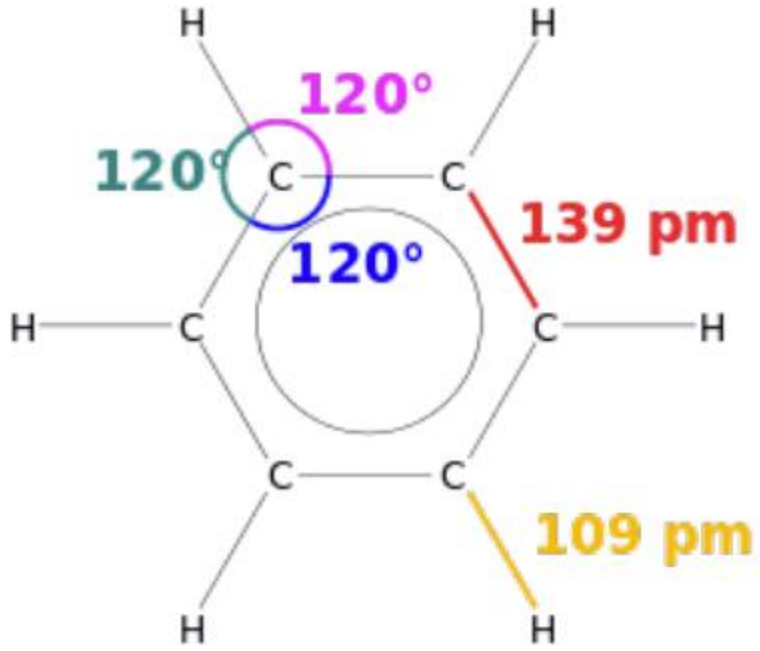
对二甲苯



Aromatic Hydrocarbons 芳香烃

C_6H_6



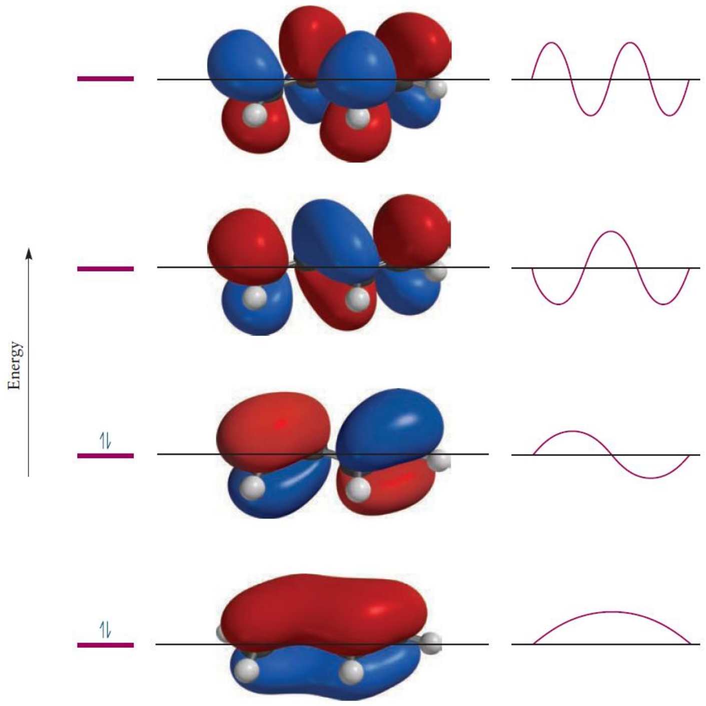


- **Benzene: 6 C-C bonds of equal length with properties intermediate between those of single and double bonds.**

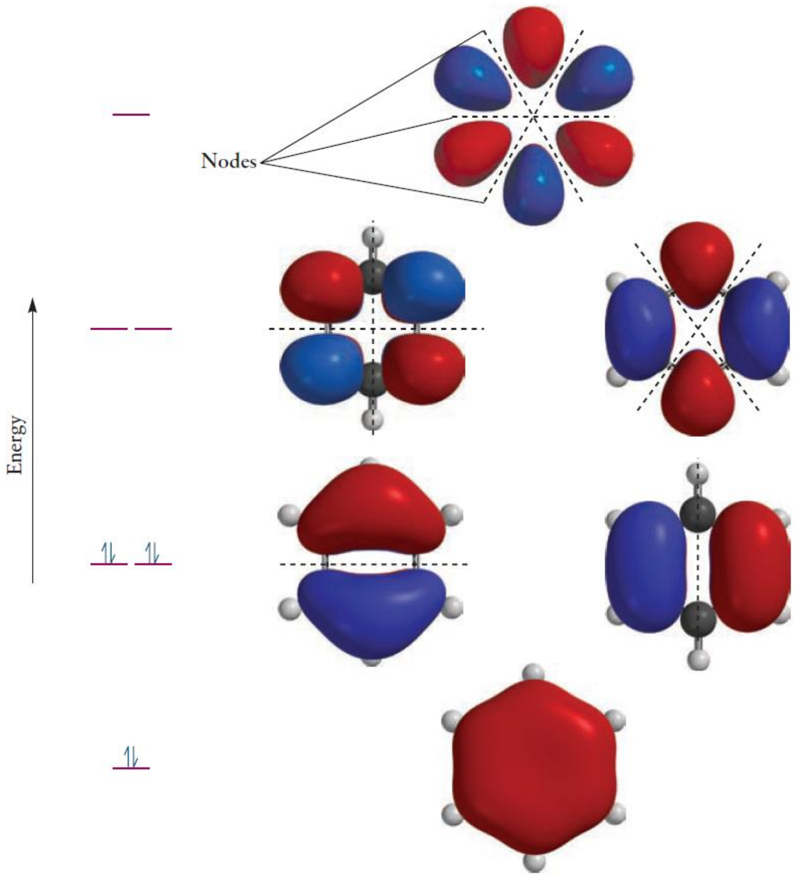
- **The modern view of resonant structures: molecule does not jump between structures, but rather has a single, time-independent electron distribution in which the π bonding is described by delocalized MOs.**

Aromatic Hydrocarbons

1. Compatible orbital symmetries
2. Maximum orbital overlap
3. Approximate atomic orbital energies



4 orbital, 4 electrons



6 orbital, 6 electrons

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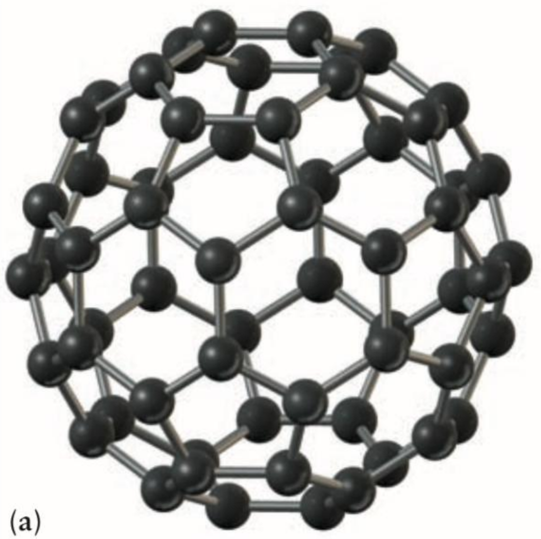
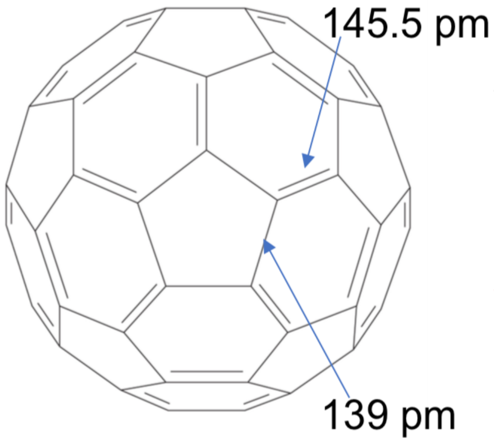
Fullerenes富勒烯



(a)

(b)



- The C_{60} molecule belongs to the I_h point group and connected in 5- and 6-membered rings;
- There are two types of C-C bond: those at the junctions of two hexagonal rings (6,6-edges), while those between a hexagonal and a pentagonal ring (5,6-edges).

The Nobel Prize in Chemistry 1996

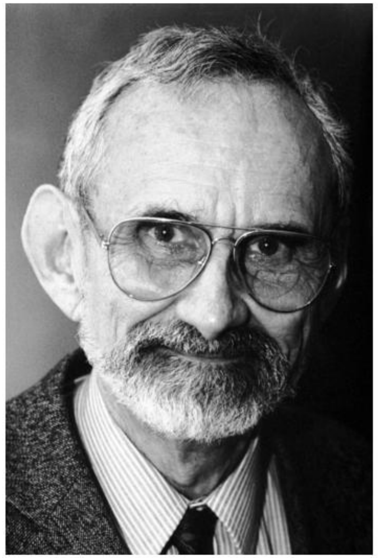
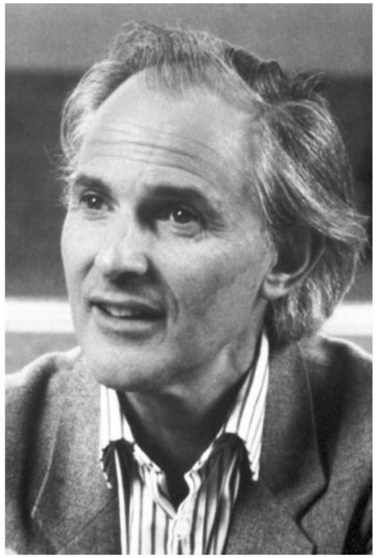


Photo from the Nobel Foundation archive.

Robert F. Curl Jr.

Prize share: 1/3



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Sir Harold W. Kroto

Prize share: 1/3

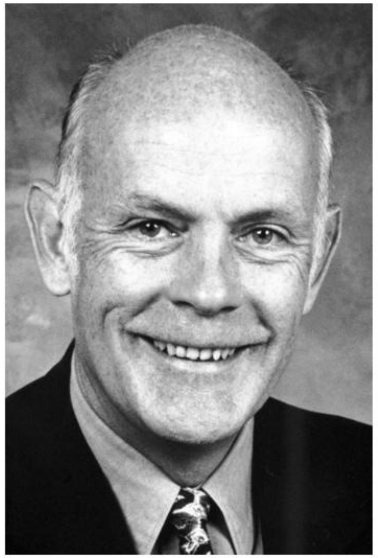


Photo from the Nobel Foundation archive.

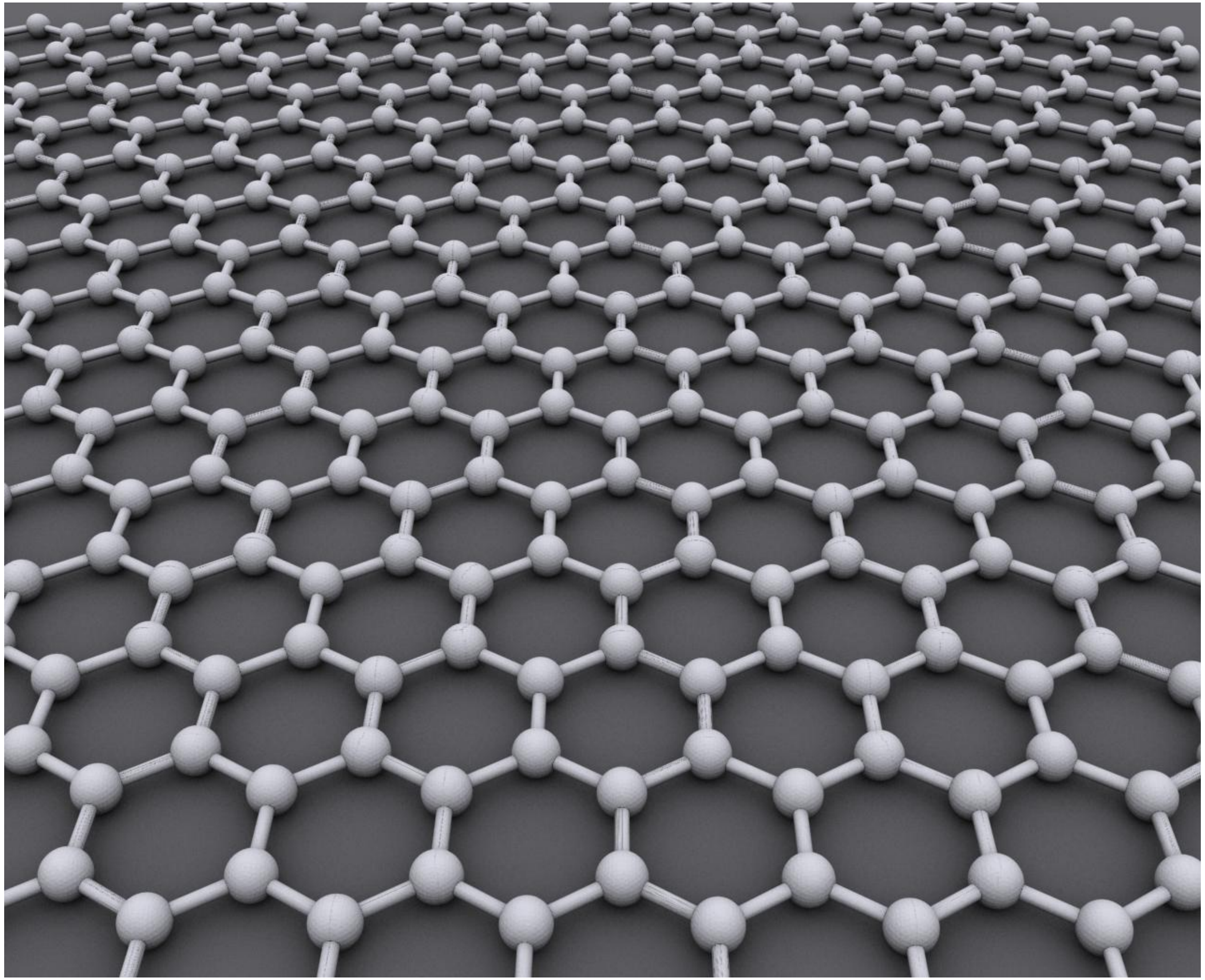
Richard E. Smalley

Prize share: 1/3

The Nobel Prize in Chemistry 1996 was awarded jointly to Robert F. Curl Jr., Sir Harold W. Kroto and Richard E. Smalley "for their discovery of fullerenes"

https://www.nobelprize.org/prizes/chemistry/1996/summary/

Graphene石墨烯



The Nobel Prize in Physics 2010

© The Nobel Foundation. Photo: U. Montan

Andre Geim

Prize share: 1/2

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Konstantin Novoselov

Prize share: 1/2

The Nobel Prize in Physics 2010 was awarded jointly to Andre Geim and Konstantin Novoselov "for groundbreaking experiments regarding the two-dimensional material graphene"

https://www.nobelprize.org/prizes/physics/2010/summary/

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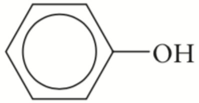
Functional Groups (官能团) and Organic Reactions

TABLE 7.3 Common Functional Groups

Functional Group†	Type of Compound	Examples
R—F, —Cl, —Br, —I	Alkyl or aryl halide	CH_3CH_2Br (bromoethane)
R—OH	Alcohol	CH_3CH_2OH (ethanol)
	Phenol	 (phenol)
R—O—R′	Ether	$H_3C—O—CH_3$ (dimethyl ether)
R—C(=O)—H	Aldehyde	$CH_3CH_2CH_2—C(=O)—H$ (butyraldehyde, or butanal)
R(R′)C=O	Ketone	$H_3C—C(=O)—CH_3$ (propanone, or acetone)
R—C(=O)—OH	Carboxylic acid	CH_3COOH (acetic acid, or ethanoic acid)
R—C(=O)—O—R′	Ester	$H_3C—C(=O)—O—CH_3$ (methyl acetate)
R—NH_2	Amine	CH_3NH_2 (methylamine)
R—C(=O)—N(R′)R″	Amide	$H_3C—C(=O)—NH_2$ (acetamide)

Functional groups: the **reactive sites**, and their chemical properties depend only weakly on the natures of the hydrocarbons.

Halides（卤化物）

The Cl atom forms a σ bond to a C atom by overlap of its $3p_z$ orbital with a hybridized orbital on the carbon (sp^3, sp^2, or sp depending on the bonding in the hydrocarbon frame).

$$\underset{\text{Methane}}{CH_4} + \underset{\text{Chlorine}}{Cl_2} \xrightarrow{250°C–400°C \text{ or light}} \underset{\text{Chloromethane}}{CH_3Cl} + \underset{\text{Hydrogen chloride}}{HCl}$$

$CH_3–X$ X= F, Cl, Br, I

Halides

- A free radical (自由基) is a chemical species that contains an odd (unpaired) electron.
- Usually formed by breaking a covalent bond to form a pair of such species.
- Radicals are often highly reactive and appear as intermediates in reactions.

Ultraviolet light initiates the reaction by dissociating a small number of chlorine molecules into highly reactive atoms.

free radical chain reaction

$$Cl\cdot + CH_4 \longrightarrow HCl + \cdot CH_3 \qquad \text{(propagation)}$$

$$\cdot CH_3 + Cl_2 \longrightarrow CH_3Cl + Cl\cdot \qquad \text{(propagation)}$$

Halides

Adding Cl to C=C bonds is a more important industrial route to alkyl halides than the free-radical reactions just described.

$$CH_2{=}CH_2 + Cl_2 \longrightarrow ClCH_2CH_2Cl$$

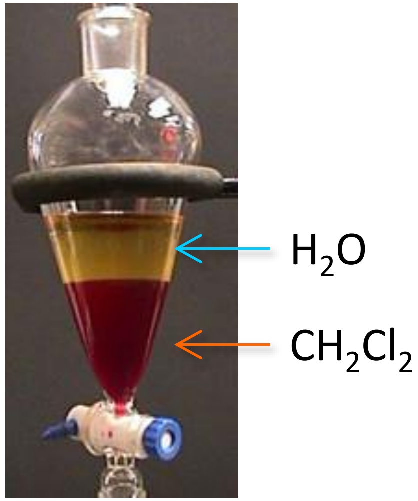
Heating the 1,2-dichloroethane to 500°C over a charcoal （木炭）catalyst to abstract HCl

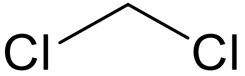
$$ClCH_2CH_2Cl \longrightarrow CH_2{=}CHCl + HCl$$

Halides

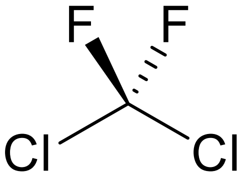
Average Bond Dissociation Energies ($kJ \cdot mol^{-1}$)

C—H	C—F	C—Cl	C—Br	C—I
414	485	339	285	213

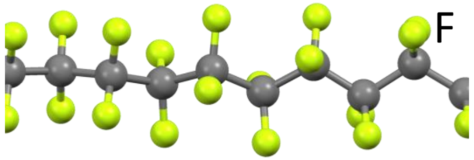




Dichloromethane
二氯甲烷



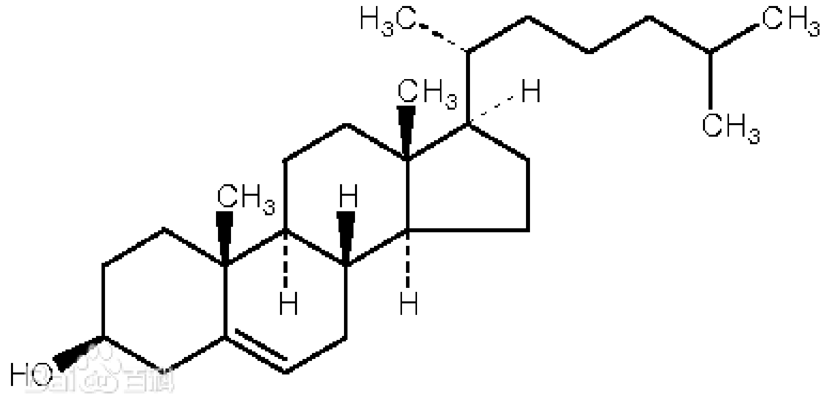
Dichlorodifluoromethane
二氟二氯甲烷
(Freon 氟利昂)



Polytetrafluoroethylene
聚四氟乙烯
(Teflon 特氟龙)

Alcohols醇类

Alcohols have the -OH functional group attached to a tetrahedral carbon atom



The simplest alcohol is methanol (CH_3OH), which is made from methane in a two-step process.

$$CH_4(g) + H_2O(g) \longrightarrow CO(g) + 3\ H_2(g)$$

$$CO(g) + 2\ H_2(g) \longrightarrow CH_3OH(g)$$

Alcohols醇类

C_2H_5OH

$$CH_2{=}CH_2 + H_2O \longrightarrow CH_3CH_2OH$$

$CH_3CH_2CH_2OH$

1-Propanol

$$\begin{array}{c} CH_3\underset{\substack{|\\ OH}}{C}HCH_3 \end{array}$$

2-Propanol

Isopropyl alcohol is made from propylene by an hydration reaction catalyzed by sulfuric acid.

$$H_3C{-}CH{=}CH_2 + H^+ \longrightarrow H_3C{-}\overset{+}{C}H{-}CH_3$$

$$H_3C{-}\underset{\substack{|\\ {}^{+}OH_2}}{CH}{-}CH_3$$

$$H_2O + H_3C{-}\underset{\substack{|\\ {}^{+}OH_2}}{CH}{-}CH_3 \longrightarrow H_3C{-}\underset{\substack{|\\ OH}}{CH}{-}CH_3 + H_2SO_4$$

Alcohols醇类

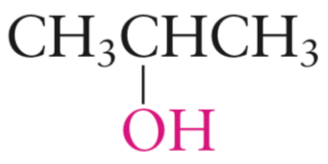
primary alcohol

$CH_3CH_2CH_2OH$

1-Propanol

沸点（℃）： 97.1

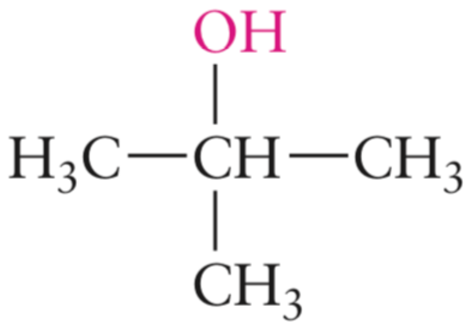
secondary alcohol



2-Propanol

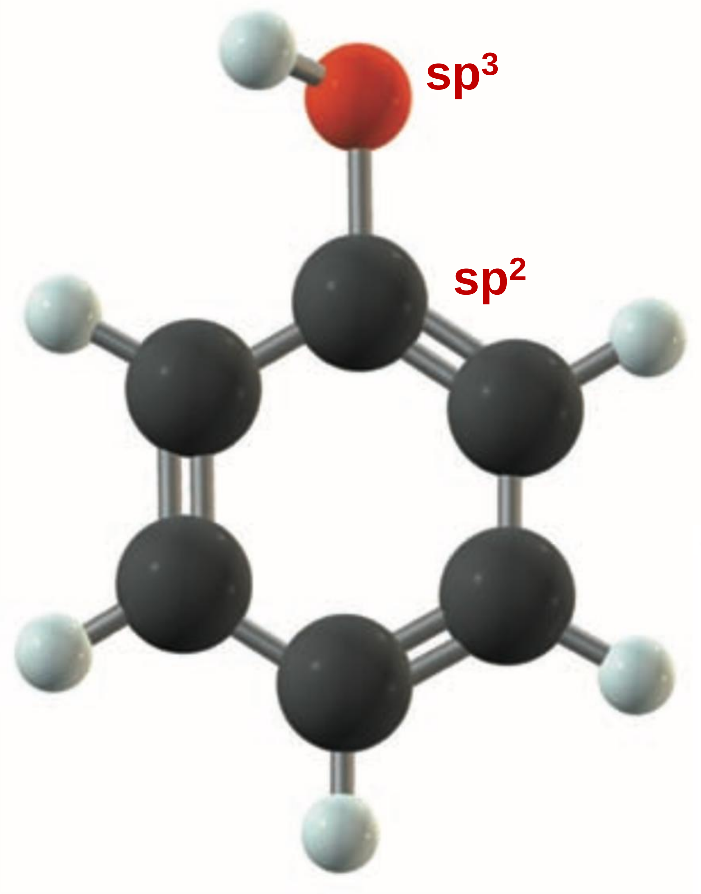
沸点：82.45

tertiary alcohol



Phenols苯酚类

Phenols （苯酚类）are compounds in which an OH group is attached directly to an aromatic ring

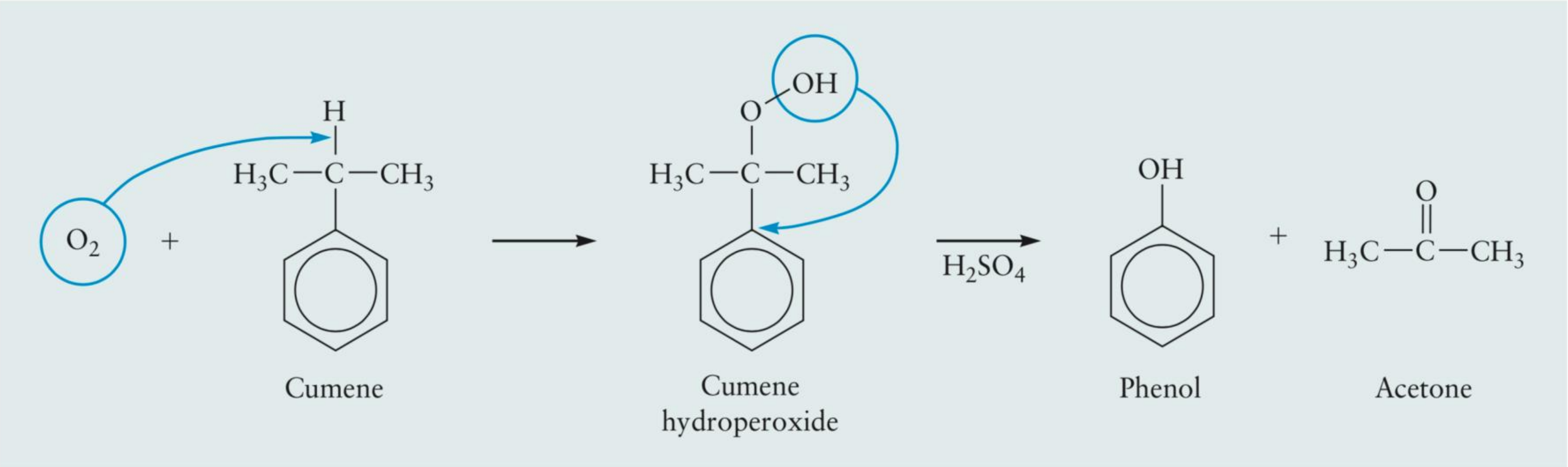


The structure of phenol, C_6H_5OH.

$$C_6H_6 + Cl_2 \longrightarrow HCl + C_6H_5Cl$$

$$C_6H_5Cl + 2\ NaOH \longrightarrow C_6H_5O^-Na^+ + NaCl + H_2O$$

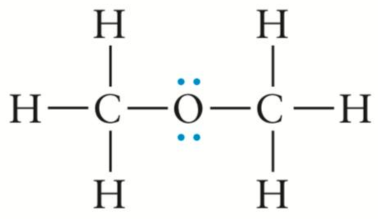
- **When chlorine reacts with an alkene, it adds across the double bond.**
- **When an aromatic ring is involved, substitution of chlorine for hydrogen.**



Ethers醚类

Ethers are characterized by the −O− functional group, in which an oxygen atom provides a link between two separate alkyl or aromatic groups.

(a)



(b) sp^3

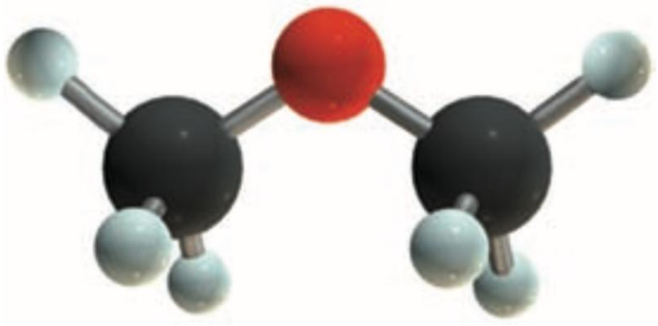
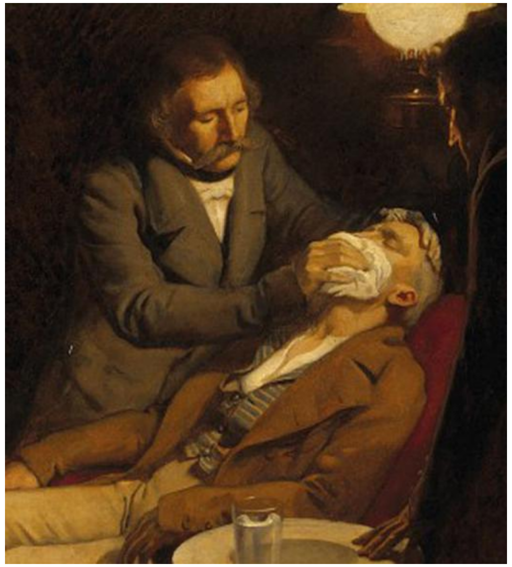


FIGURE 7.24 Structure of dimethyl ether. (a) Lewis diagram. (b) Ball-and-stick model.

It can be produced by a condensation reaction.

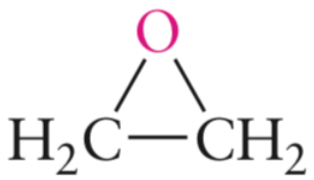
$$CH_3CH_2OH + HOCH_2CH_3 \xrightarrow{H_2SO_4} CH_3CH_2{-}O{-}CH_2CH_3 + H_2O$$



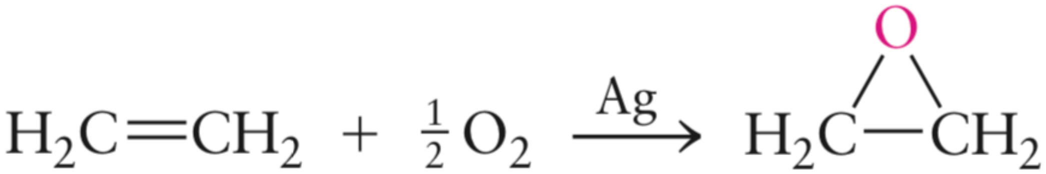
Anesthesia 麻醉术

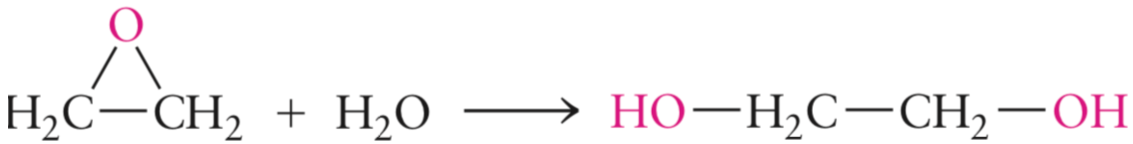
Ethers醚类

Cyclic ether: oxygen forms part of a ring with carbon atoms



epoxides 环氧化物





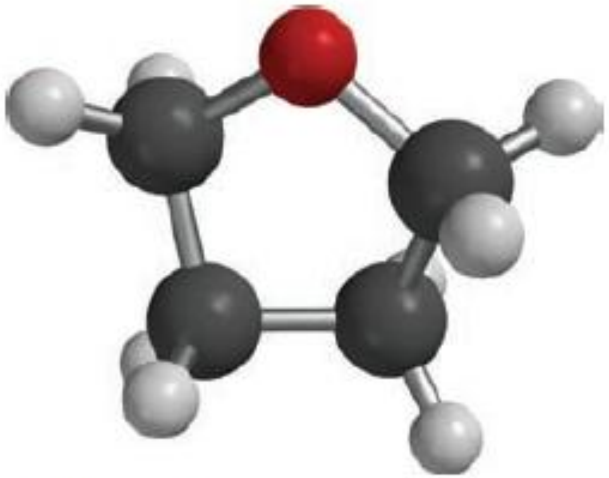
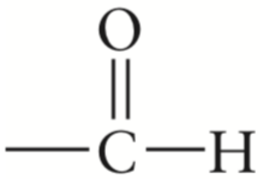
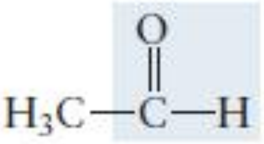


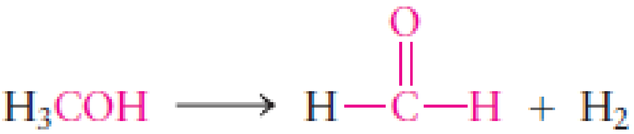
FIGURE 7.22 The structure of tetrahydrofuran, C_4H_8O.

Aldehydes醛类

An aldehyde contains the characteristic functional group  **in its molecules**



Acetaldehyde
(an aldehyde)



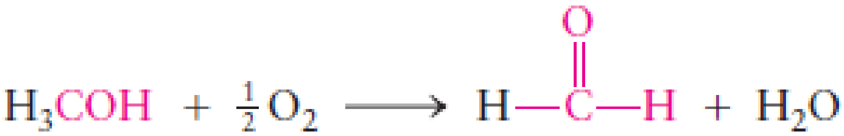
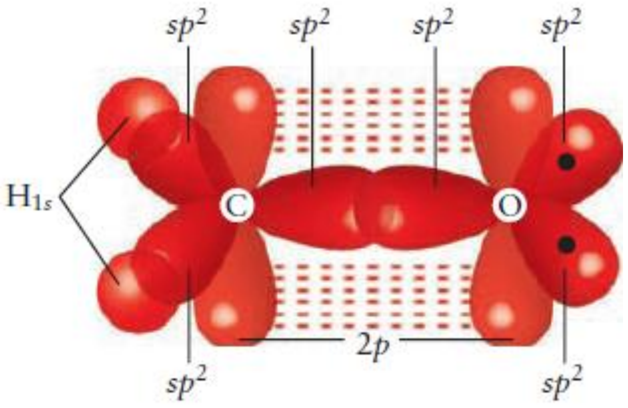
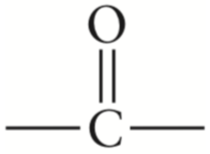


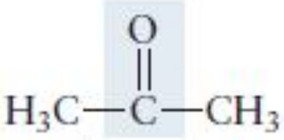
FIGURE 7.23 Bonding in formaldehyde involves sp^2 hybrid orbitals of both the carbon (C) and oxygen (O) atoms. The C—O σ bond is constructed by overlap of C and O sp^2 hybrid orbitals and the C—H bonds from overlap of C sp^2 orbitals with H 1s orbitals. Two lone pairs occupy the other O sp^2 hybrid orbitals. The π bond is formed by overlap of the parallel unhybridized 2p orbitals.



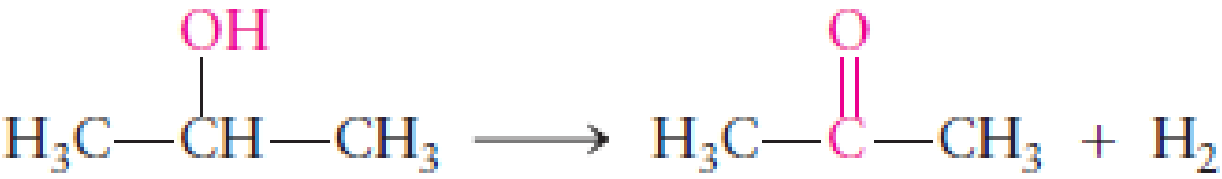
Both C and O are sp^2 hybridized.

Ketones酮类

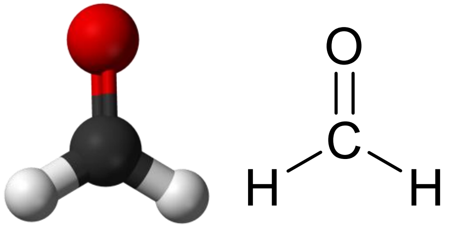
Ketones have the  **functional group in which a carbon atom forms a double bond to an oxygen atom and single bonds to two separate alkyl or aromatic groups**



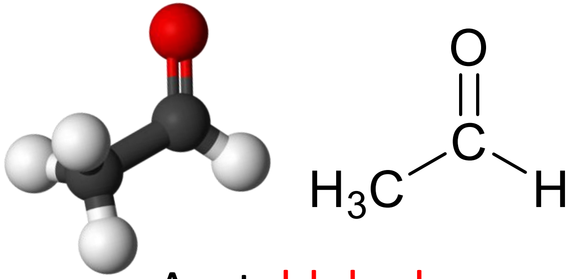
Acetone
(a ketone)



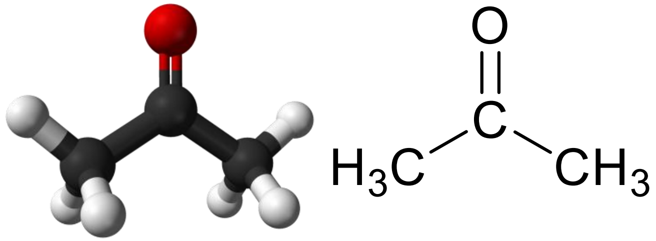
Aldehydes and Ketones



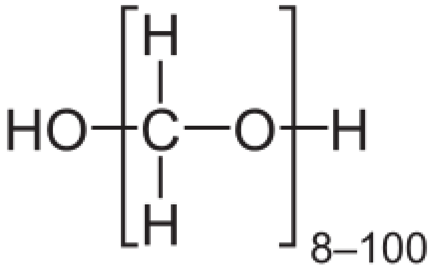
Formaldehyde
Methanal
甲醛



Acetaldehyde
Ethanal
乙醛

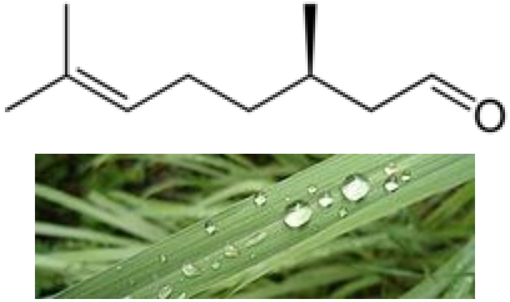


Acetone
Propanone
丙酮



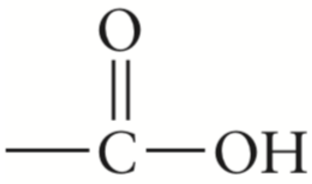
Paraformaldehyde
多聚甲醛

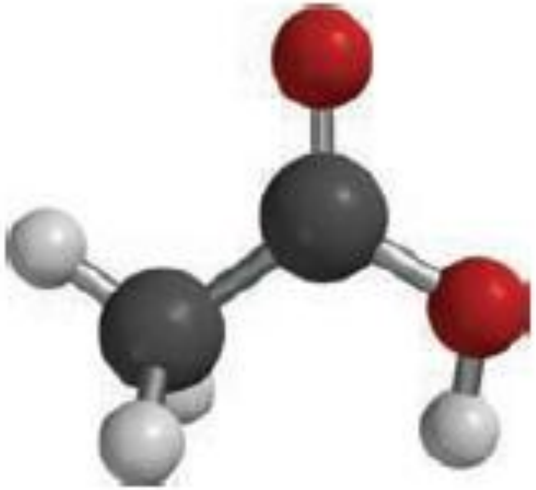
Cinnamaldehyde
肉桂醛

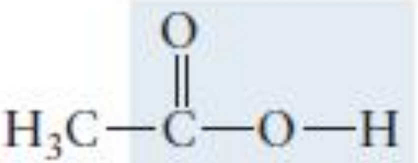


Citronellal
香茅醛

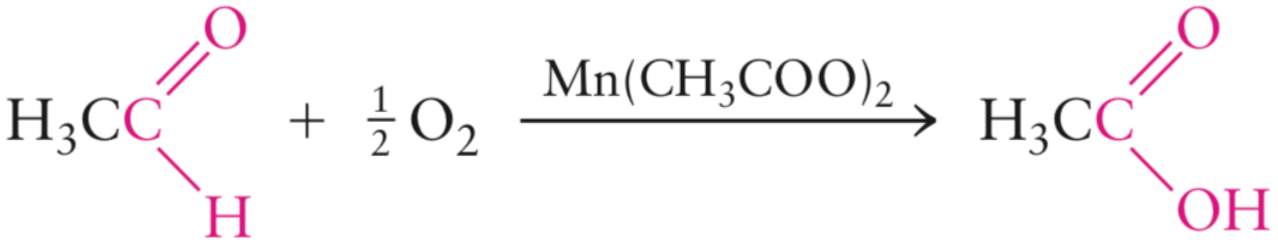
Carboxylic Acids羧酸

Carboxylic acids contain the  **functional group (also written as -COOH).**





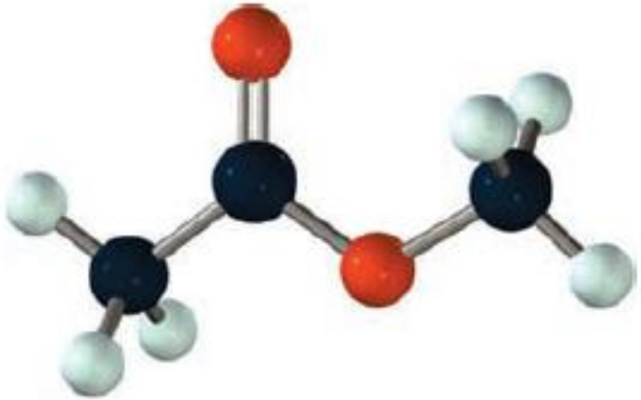
Acetic acid
(a carboxylic acid)

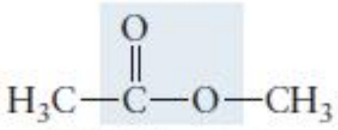


$$CH_3OH(g) + CO(g) \xrightarrow{Rh,I_2} CH_3COOH(g)$$

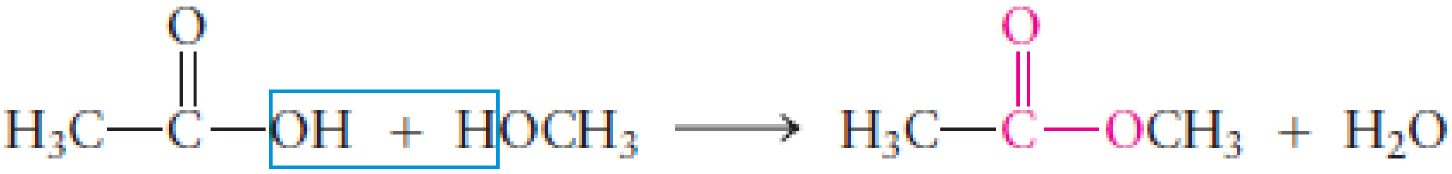
Esters羧酸酯

Carboxylic acids react with alcohols or phenols to give esters, forming water as the coproduct



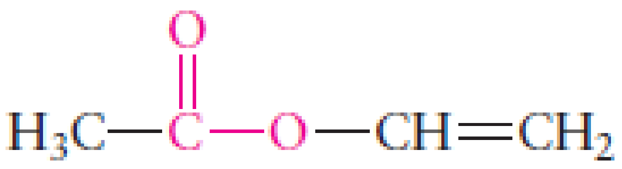


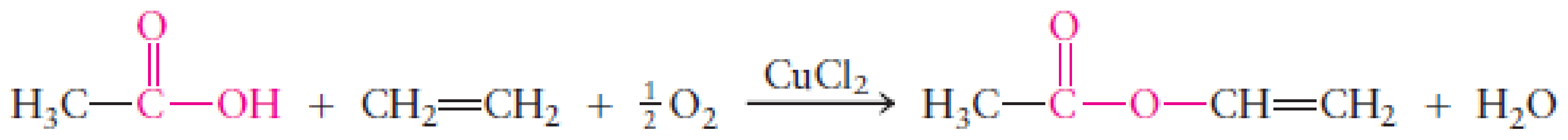
Methyl acetate
(an ester)



Esters羧酸酯

Esters are named by stating the name of the alkyl group of the alcohol (the methyl group in this case), followed by the name of the carboxylic acid with the ending -ate (acetate).



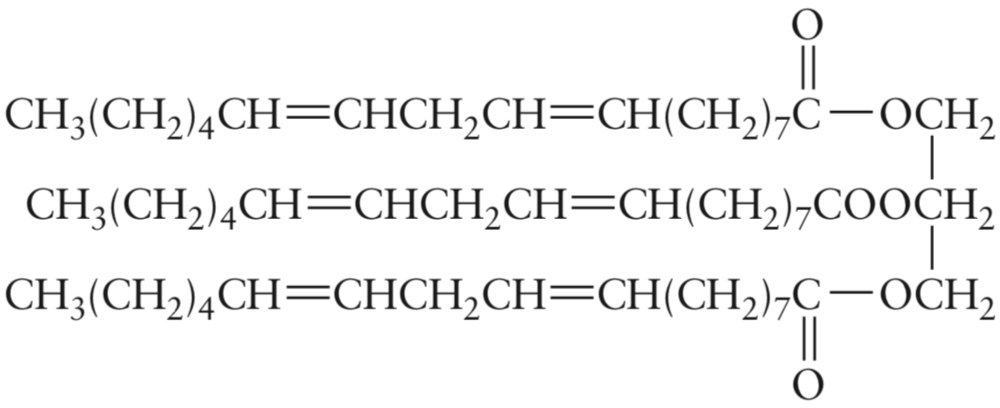


Esters are colorless, volatile liquids that often have pleasant odors.

Many occur naturally in flowers and fruits.

Esters羧酸酯

<u>Animal fats</u> and <u>vegetable oils</u> are tri-esters of long-chain carboxylic acids with glycerol （甘油，丙三醇）, $HOCH_2CH(OH)CH_2OH$, a tri-alcohol; they are referred to as triglycerides.



This is a polyunsaturated oil because there are six C=C bonds per molecule.

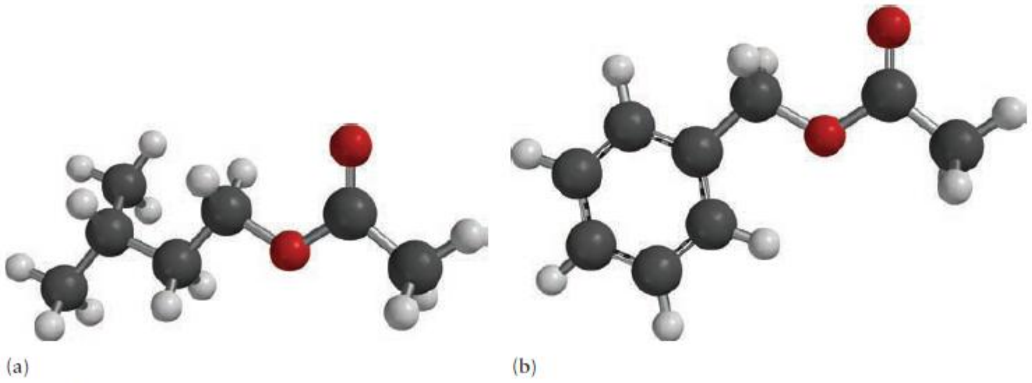
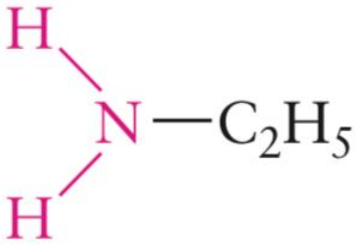
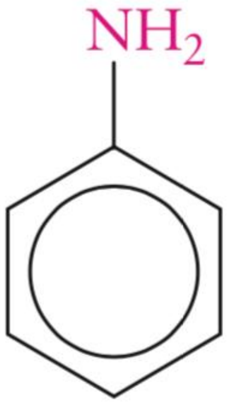
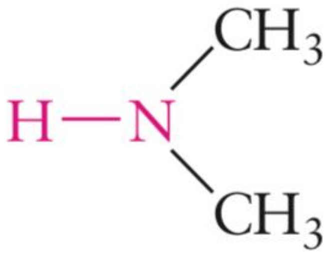
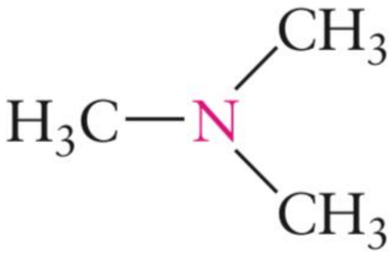


FIGURE 7.24 The structures of (a) isoamyl acetate, $CH_3COO(CH_2)_2CH(CH_3)_2$, and (b) benzyl acetate, $CH_3COOCH_2C_6H_5$.

Amines 胺类

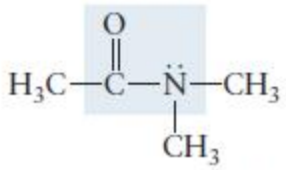
The amines are derivatives of ammonia with the general formula R_3N, where R can represent a hydrocarbon group or hydrogen.

			
Ethylamine	Aniline	Dimethylamine	Trimethylamine
primary amine	**primary amine**	**secondary amine**	**tertiary amine**

N is sp^3 hybridized

Amides 酰胺

A primary or secondary amine (or ammonia itself) can react with a carboxylic acid to form an amide



Dimethylacetamide
(an amide)

$$H_3C-\overset{\displaystyle O}{\overset{\|}{C}}-\boxed{OH\ +\ H}-N(CH_3)_2 \longrightarrow H_3C-\overset{\displaystyle O}{\overset{\|}{C}}-N(CH_3)_2 + H_2O$$

$$H_3C-\overset{\displaystyle O}{\overset{\|}{C}}-OH + NH_3 \longrightarrow H_3C-\overset{\displaystyle O}{\overset{\|}{C}}-NH_2 + H_2O$$

FIGURE 7.25 Bonding in dimethyl acetamide, an amide. The amide linkage is planar.

Amide linkages are present in the backbone of every protein molecule and are very important in biochemistry, where the structure of the molecule strongly influences its function

Homework: read "principle of modern chemistry", chapter 7